

Secure and Privacy-Preserving Intrusion Detection in Wireless Sensor Networks via Perturbation Adversarial Training

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Background: WSNs & Intrusion Detection

- **Wireless Sensor Networks (WSNs):**

- Critical component of modern cyber-physical systems (Industrial automation, smart cities, healthcare).
- Consists of distributed nodes with severe constraints in computational power, memory, and energy.
- **Vulnerable to a various cyber and physical attacks** (e.g., sinkhole, Sybil, jamming).

- **Machine Learning-based IDS:**

- ML and deep learning have emerged as promising tools for detecting anomalies and attacks.
- **Vulnerability:** ML-based IDS models are highly **susceptible to adversarial attacks** where inputs are carefully perturbed.

Motivation: The Threat Landscape

Traditional Adversarial Defenses:

- Focus exclusively on **digital domain perturbations** of network traffic features at higher protocol layers.

The WSN Reality (Cyber-Physical):

- Attackers can simultaneously exploit *both*:
 - ① **Digital attack** (manipulate routing packets or traffic statistics)
 - ② **Physical-layer attack** (alter transmission power, timing, or channel characteristics)

Hybrid Cyber-Physical Adversarial Attacks

Combine subtle digital traffic perturbations with physical-layer distortions to **evade detection stealthily**.

*Such attacks cause false negatives or false positives, **eroding trust** and endangering critical infrastructure. Current IDS training pipelines do not model such hybrid threats.*

Problem Formulation

Two perturbation sources threaten model robustness:

① Digital-level adversarial attacks (A_ϵ):

- Subtle manipulations of traffic features: $X_\epsilon = A_\epsilon(X)$, bounded by $\|X_\epsilon - X\|_p \leq \epsilon$

② Physical-layer transformed perturbations (T_s):

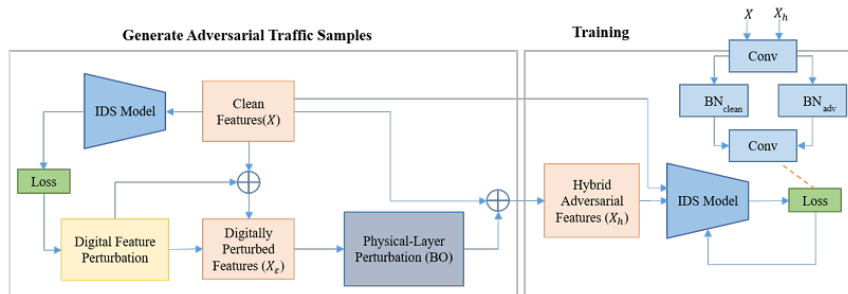
- $T_s(X) = \alpha \cdot X + \beta \cdot \frac{\partial X}{\partial t} + \mathcal{N}(0, \tau)$
- $s = (\alpha, \beta, \tau)$: amplitude scaling (α), timing/frequency drift (β), and additive channel noise (τ).

Combined Hybrid Adversarial Attack

$$X_h = T_s(A_\epsilon(X))$$

Chains these two threats, representing the most realistic adversarial condition.

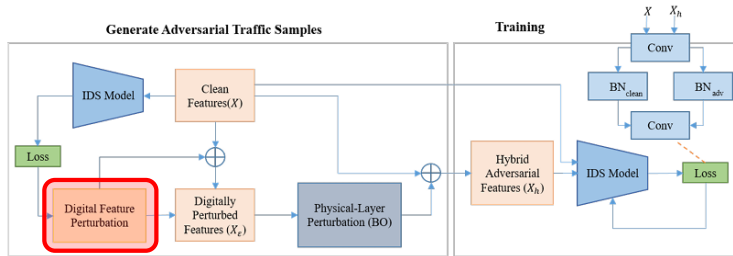
Proposed Framework: Overview



Three Integrated Components:

- **PGD:** Generates adversarial perturbations in the digital feature space.
- **BO:** Efficiently searches for worst-case physical-layer attack parameters.
- **MixBN:** Mitigates performance degradation on benign (clean) data.

1. Digital Feature Perturbations (PGD)

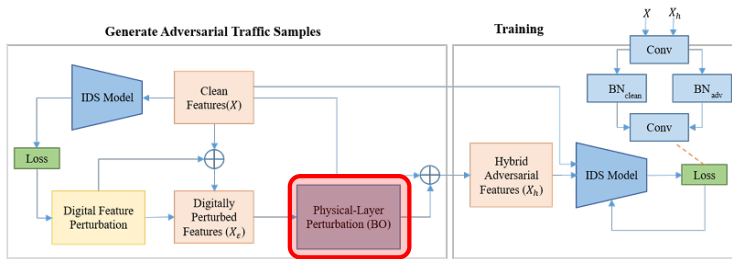


- **Goal:** Simulate adversarial manipulations in the digital domain (e.g., packet forging).
- Iteratively updates input to maximize loss under a bounded constraint:

PGD Update Rule

$$X_\epsilon^{t+1} = \Pi_\epsilon \left(X_\epsilon^t + \epsilon \cdot \text{sign} \left(\frac{\nabla_X \ell(f_\theta(X_\epsilon^t), y)}{\|\nabla_X \ell(f_\theta(X_\epsilon^t), y)\|} \right) \right)$$

2. Physical-Layer Perturbations (BO)

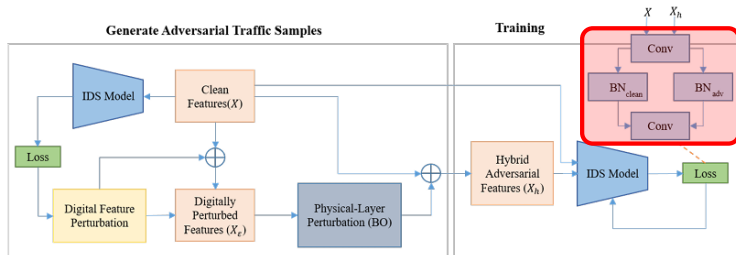


- **Goal:** Emulate physical vulnerabilities (jamming, power scaling) without exhaustive simulation.
- Uses **Bayesian Optimization** (GP + Expected Improvement) to find worst-case s :

BO Objective

$$s^* = \arg \max_s \mathbb{E}_{(X,y) \sim \mathcal{D}} [\ell(f_\theta(T_s(A_\epsilon(X))), y)]$$

3. Mixed Batch Normalization (MixBN)

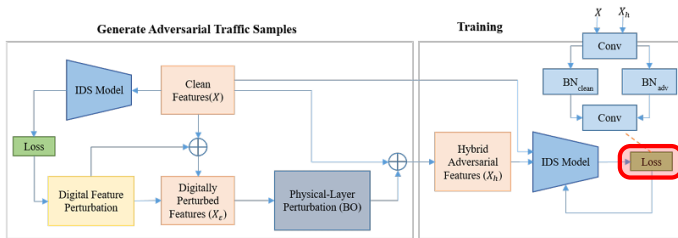


- **Dilemma:** Adversarial training degrades performance on clean data (distribution shifts).
- **Solution:** Maintain separate normalization statistics for clean and adversarial samples.

MixBN Formulation

$$\eta(X, X_h) = \text{ReLU}\left(\text{BN}_{\text{clean}}(\text{Conv}(X)) + \text{BN}_{\text{adv}}(\text{Conv}(X_h))\right)$$

Unified Training Objective



Robust Optimization Problem

$$\min_{\theta} \mathbb{E} \left[\underbrace{\ell(f_{\theta}(X), y)}_{\text{Clean Acc.}} + \underbrace{\max_{\epsilon, S} \ell(f_{\theta}(X_h), y)}_{\text{Worst-case Hybrid Attack}} + \underbrace{\eta(X, X_h)}_{\text{MixBN}} \right]$$

- Balances high accuracy on benign traffic with resilience against strong hybrid attacks.

Experimental Setup

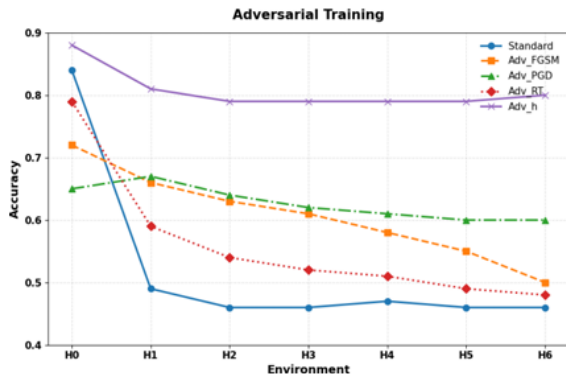
Dataset Description:

- **WSN-DS (Dataset 1):** 374,661 records, 19 features. Includes Blackhole, Grayhole, Flooding, and TDMA scheduling attacks.
- **CIC-IDS 2017 (Dataset 2):** 2,496,897 records. Comprehensive benchmark covering a wide spectrum of intrusion scenarios (DDoS, brute-force, infiltration).

Adversarial Environments ($H_0 \sim H_6$):

- H_0 : Clean network environment.
- $H_1 \sim H_6$: Progressively intensified hybrid adversarial environments.
- Incremental increases in digital (ϵ) and physical (α, β, τ) perturbation strengths.

Results: Robustness under Hybrid Attacks

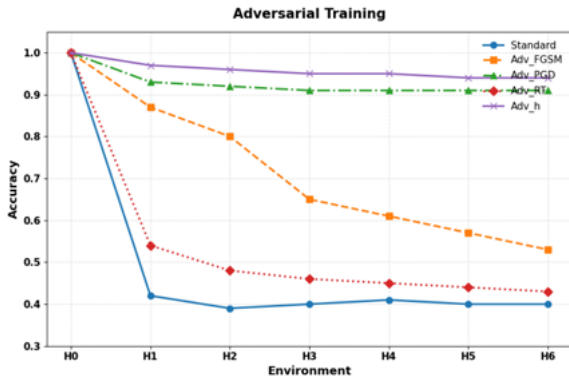


Legend Key:

- **Standard**: Clean-data only
- **Adv_FGSM**: Single-step digital defense
- **Adv_PGD**: Iterative digital-only defense
- **Adv_RT**: Physical-layer only defense
- **Adv_h**: **Proposed Hybrid Defense**

Adv_h maintains high accuracy across $H_1 \sim H_6$.

Results: Rate of Invariance (ROI)



Stability Comparison:

- **Standard:** ROI drops below 0.5.
- **Adv_h:** ROI consistently > 0.95 .

Majority of samples retain consistent predictions even under severe attacks.

Results: Ablation Study on MixBN

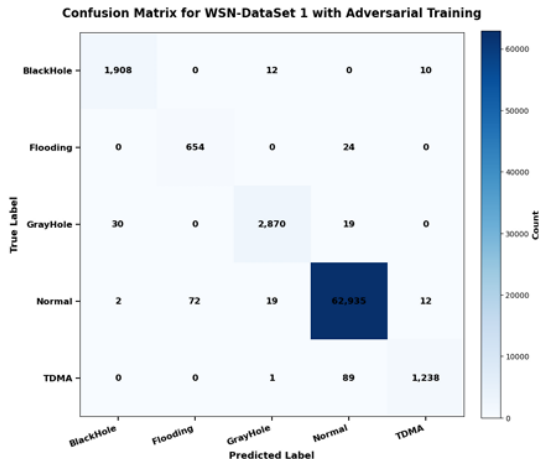
Introducing MixBN consistently improves accuracy, precision, F1, and AUC.

Model	ACC	PRE	F1	AUC
Adv_p	69.25	71.84	75.39	61.35
$\text{Adv}_p + \text{MixBN}$	71.54	74.41	79.15	73.87
Adv_{RT}	81.47	80.37	87.36	88.39
$\text{Adv}_{RT} + \text{MixBN}$	82.74	85.74	87.51	89.84
Adv_h	76.12	76.38	80.37	77.95
$\text{Adv}_h + \text{MixBN}$	78.92	76.32	823.63	81.37

Figure: Ablation of MixBN across training schemes

Validates that normalizing clean and hybrid-adversarial streams with separate statistics mitigates distribution shifts.

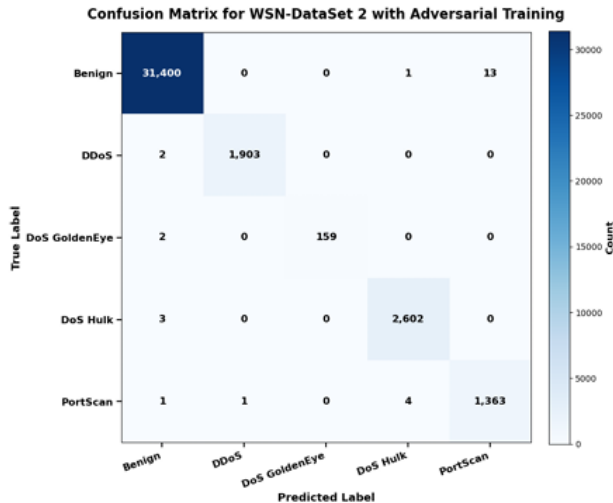
Results: Confusion Matrix (WSN-DS)



Dataset 1 Analysis:

- **Normal:** 62,935 samples correctly classified.
- **Attacks:** High precision for BlackHole (1,908) and GrayHole (2,870).
- **Precision:** Misclassifications represent $< 0.5\%$ of total samples.

Results: Confusion Matrix (CIC-IDS 2017)



Dataset 2 Analysis:

- **Scale:** Evaluated on 2.5M records benchmark.
- **Attacks:** Exceptional accuracy for DDoS (1,903) and DoS-Hulk (2,602).

Conclusion & Future Work

Conclusion:

- Proposed a **Perturbation Adversarial Training framework** integrating PGD, BO, and MixBN.
- Effectively enhanced model robustness against hybrid cyber-physical threats.
- Consistently outperforms baseline approaches (Adv_{FGSM} , Adv_{PGD} , Adv_{RT}), maintaining higher accuracy and ROI.

Future Work:

- Explore **adaptive adversarial training**.
- Develop **lightweight, federated variants** to enhance scalability and real-time deployment in large-scale WSNs.

Thank you!

Any Questions?

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